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EXPERIMENT - 7

Ques :- Given n cities with flights having [from, to, price]. Given integers src, dst, k. Return cheapest price from src to dst with atmost k stops. Return -1 if no such route exists.

Approach :- start from src with cost = 0 and stops = 0 at every step. Explore neighbouring flights and update minimum cost to reach each node.

Only continue exploring if stops ≤ k. Update the distance of node if finds a cheapest node.

Pseudo code :

```
// create Adjacency Matrix
for (auto x : flights) {
    adj [x[0]] . push_back ({ x[1], x[2]});
}

queue <pair <int, pair <int, int >>> q;
// stops, node, dist

vector <int > dist (n, 1e9);

q.push ({0, {src, 0}});
```

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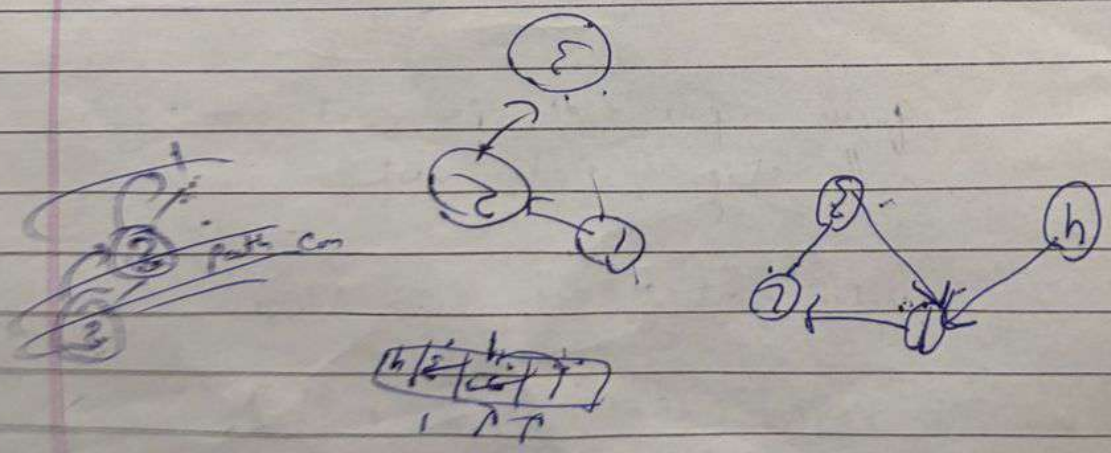
```
while (!q.empty()) {
    int stops = q.front().first;
    int node = q.front().second.first;
    int dist = q.front().second.second;
    q.pop();

    if (stops > k) continue;

    for (auto x : adj[node]) {
        int v = it.first;
        int wt = it.second;
        if (cost + wt < dist[v]) {
            dist[v] = cost + wt;
            q.push({stops + 1, {v, dist[v]}})
        }
    }
}
if (dist[dst] == 1e9) return -1;
return dist[dst];
}
```

TC : O(E * K) SC : O(V + E)